

MKUFT

A Relational Architecture for Physical Law and Cross-Scale Dynamics

Mark Charles McLaughlin

Independent Researcher

ORCID: [0009-0005-7736-1511](https://orcid.org/0009-0005-7736-1511)

Publication date	17 August 2026
Version DOI	10.5281/zenodo.21973064
Concept DOI	10.5281/zenodo.17780565
Historical DOI	10.5281/zenodo.17780566
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0)
Repository	Public MKUFT canon

Publication boundary. This paper is the current principal publication in the historical MKUFT concept lineage. It is a frozen standalone publication reconciled against the public canon. It does not alter the historical publication ([10.5281/zenodo.17780566](https://doi.org/10.5281/zenodo.17780566)), and later repository revisions will not be silently inserted into this publication.

Abstract

The McLaughlin–Kairos Unified Field Theory (MKUFT) is an evolving research programme for analysing how possibility, relational constraint, physical dynamics, measurement, history, and scale-specific description fit together. Its present contribution is not a completed unified field theory and does not include a derivation of quantum mechanics, quantum field theory, general relativity, or quantum gravity. It is an addressed architecture for deciding which objects and relations are admissible before an effective law is selected, and for testing whether a proposed higher-level description genuinely owns the dynamics attributed to it.

The framework begins with a typed substrate–information–physical–observer structure, S–I–P–O, and replaces an earlier multiplicative probability ansatz with a more general factorisation. A complete addressed state and measured context determine an admissibility descriptor; a law assembler constructs the active physical-law object—its domain, transition rule, and any domain-appropriate weighting; physical propagation and a typed measurement instrument produce outcomes and records; the realised path then readdresses the next effective possibility object. Relational organisation is treated as a cross-implementation functional invariant only when a declared relation survives representation change, carrier substitution, and deformation while continuing to perform the claimed work. Across scale, physical persistence, independent addressability, functional identity, constitutive closure, law sufficiency, capability, cost, and recoverability are kept as different objects connected by explicit comparison maps.

The mathematical objects are classified as notation, research scaffolds, operational models, or recovery obligations rather than presented uniformly as laws of nature. Bell compatibility, no-signalling, Born-rule and quantum-field-theory recovery, gravitational recovery, conservation, and complete energy accounting are hard acceptance tests for any foundational physical instantiation. A strongest-fair-null programme, controlled deformation, complete-history replay, held-out prediction, and branch-specific falsifiers define how the architecture can be reduced or rejected. The result is the current principal MKUFT publication: a self-contained statement of the research architecture, its mathematics, its experimental programme, and the boundary between what is organised and what remains scientifically unearned.

Keywords: MKUFT; Layer Before Law; SIPO architecture; relational admissibility; effective physical law; addressed state space; law descent; cross-scale dynamics; recoverability; quantum foundations; quantum gravity; falsifiability.

1. Publication identity, lineage, and status

This paper is the current principal MKUFT publication in the historical concept lineage established by *Unified Field Theory. McLaughlin–Kairos. MKUFT*, DOI [10.5281/zenodo.17780566](https://doi.org/10.5281/zenodo.17780566) and concept DOI [10.5281/zenodo.17780565](https://doi.org/10.5281/zenodo.17780565). The historical object remains unchanged and citable. It records the early S–I–P–O proposal, observer-linked probability scaffolds, exploratory experiments, and falsification intentions. It also contains mathematical and typographic limitations that the present publication does not conceal.

The present paper is reconciled against the current public MKUFT GitHub canon. It is not a concatenation of that repository. The repository remains the evolving supporting corpus; this paper is a frozen standalone synthesis with its own publication identity.

Two related standalone papers remain distinct objects. *Layer Before Law* develops the addressed admissibility and quantum-gravitational recovery proposal in focused form [20]. *Recursive Constraint Closure and Reachable-State Geometry* develops adaptive state-space and endogenous-admissibility machinery [21]. This publication integrates the parts of those results required for a current statement of MKUFT, without silently replacing either paper.

The scientific status is deliberately narrow:

- MKUFT is an original layered research architecture using mostly established mathematical tools.
- Its current equations include notation, consistency conditions, research scaffolds, and operational tests.
- No current foundational MKUFT equation is a demonstrated new law of nature.
- No independent physical effect has yet established an S-layer or I-layer cause beyond an adequate physical-state, history, boundary, environment, and measurement model.
- A foundational MKUFT branch succeeds only if it recovers established physics and produces a discriminating result beyond strong alternatives.

1.1 Substantive changes from the historical formulation

The present publication does not preserve an early formulation merely because it appeared in the historical DOI object. It makes the following load-bearing corrections.

1. The historical multiplicative probability product is no longer a governing equation of the flagship. Any mathematically repaired descendant remains an optional E1 research branch in the supporting canon. The primary architecture is now the addressed construction of a complete physical-law object followed by physical propagation, measurement, and readdressing.
2. Observer language no longer supplies a missing physical mechanism. Measurement is represented by a physical instrument that may alter P while creating an O-addressed record. Any additional observer-conditioned term must survive an adequate physical state, history, boundary, environment, and measurement-context model.
3. The substrate scaffold is separated from the effective possibility object active at a particular time. Readdressing ordinarily recomputes the latter; it does not imply that observation rewrites a universal substrate.
4. State-space address is allowed to change. Before/after subtraction, distance, state identity, and claims of preserved freedom are forbidden across different construction addresses unless a lawful comparison map has been supplied.
5. Relational organisation is no longer licensed by verbal similarity, recurring geometry, correlation, or common naming. Functional identity requires a declared load-bearing relation, a persistence or restoration criterion, lawful comparison across implementations, and deformation evidence.
6. Scale transition is decomposed into physical persistence, loss or gain of independent addressability, individuality readdressing, higher-scale capability, and law sufficiency. These events may coincide, lead or lag, or fail independently.
7. A higher-order object does not automatically own an autonomous law. Law descent is property-relative and can be exact, approximate, ensemble-conditional, history-augmented, or coupled multiscale. Evidence can require the active description to readdress upward, downward, or to a coupled scale.
8. Loss of a property-specific higher law, loss of constitutive closure, and loss of target-relative recoverability are different boundaries. Recovery is a constrained reachability claim, not a promise of reversal.
9. Strongest-fair-null, representation, carrier-substitution, history-replay, ablation, energy-accounting, and accepted-limit recovery gates replace weak comparison conditions that could manufacture apparent support.

These corrections narrow several early claims, but they also strengthen the architecture. They convert broad relational intuitions into a set of typed questions whose failure conditions can be stated before a favoured outcome is known.

2. The problem: law selection begins after object selection

Quantum theory and general relativity are highly successful in their demonstrated regimes, but their standard mathematical structures are not straightforwardly unified. One response is to search directly for a deeper common dynamical law. MKUFT asks an earlier question: before selecting or combining laws, has the effective object, its address, its admissible state domain, and the property whose evolution is being predicted been typed correctly?

The proposal is not that ordinary laws are chosen by an unconstrained observer. It is that every physical law already acts on a domain, under boundary and initial conditions, with a specified state representation and measurement route. If those inputs change across scale, history, or context, the lawful object may change even when the underlying physics remains entirely ordinary. A deeper claim—that relational or informational structure has independent physical content—must survive the strongest adequate physical description.

The architecture therefore separates six questions that are often compressed into one:

1. What is the current effective possibility object?
2. Which relational and physical assignments are admissible?
3. What physical state domain, generator, and weighting are active?
4. How is the system propagated and measured?
5. Which scale or layer owns the target property after the update?
6. What observation would force reduction to an ordinary account or rejection of the branch?

This separation is the meaning of “layer before law.” It is an architectural order of operations, not evidence for a new field.

3. Typed S–I–P–O architecture

The four-address shorthand is

$$S \longrightarrow I \longrightarrow P \longrightarrow O.$$

The symbols denote typed roles rather than four spatial dimensions.

3.1 Substrate or structured possibility

The S-address denotes an ambient possibility scaffold. A minimal mathematical representation is a measure space

$$S = (\Omega, \Sigma, \mu),$$

where Ω is a set of candidate configurations, Σ a sigma-algebra, and μ a measure. This is a representation of structured possibility, not evidence for a detected material medium. It becomes a probability space only when $\mu(\Omega) = 1$.

The ambient scaffold is separated from the effective possibility object S_t^{eff} , which contains the support currently reachable or addressable under the active construction, boundary, and history.

3.2 Information or relational constraint

The information address carries correlations, constraints, codes, relations, and organisational structure. To avoid confusing the layer label I with its mathematical carrier, define

$$\mathcal{I} = L^2(\Omega, \mu), \quad i \in \mathcal{I}.$$

An information structure does not become an independent physical cause merely because it is useful, compressive, or predictive. If $I_t = f(X_t, H_t)$ for an adequate physical state X_t and measured history H_t , then I_t may remain an effective description without establishing autonomous I→P dynamics.

3.3 Physical expression

The P-address contains measurable states, fields, matter, energy, instruments, environments, timing, and dynamics. Physical claims require physical variables, units, conservation accounting where applicable, and comparison with the strongest relevant physical baseline.

3.4 Observer-positioned registration

The O-address contains measurement context, record, attention, interpretation, memory, and bounded state variables when operationally defined. Ordinary measurement back-action remains P-layer physics. Consciousness is not inserted as a cause of quantum collapse by default, and observer language cannot replace a missing coupling.

3.5 Complete addressed state

The live state used by the capstone is

$$\mathcal{U}_t = (S_t^{\text{eff}}, I_t, P_t, O_t, b_t, H_t),$$

where b_t is the contextual or construction address and H_t is the retained load-bearing history. Measured or controlled environmental context is denoted E_t where required.

The tuple is bookkeeping until its components, comparison maps, and couplings are defined. Its purpose is to prevent a change in address, history, or measurement route from being mistaken for motion inside one unchanged state space.

4. Address, context, comparison, and functional identity

4.1 Typed address and the no-inheritance rule

For a candidate property K at layer or scale L , write

$$K_L = A_L(K; \theta_L),$$

where θ_L collects the variables, constraints, observables, units, tolerances, and admissible transitions of that address. A cross-address coupling has typed source and target spaces,

$$C_{LM} : \mathcal{X}_L \longrightarrow \mathcal{X}_M.$$

Repeated algebraic form is not a coupling. A lawful cross-layer claim identifies the relation preserved, the translation or coarse-graining used, the evidence appropriate to the receiving layer, and an independent falsifier.

No untyped inheritance. A property may cross a scale or layer boundary only when the relation, address, and evidence preserve the property required by the downstream claim.

The rule applies to causation, evidence, units, identity, functional role, and law ownership. A relation that carries lineage need not carry functional identity; a relation that carries functional organisation need not carry token identity; and a coarse-graining that carries a passive forecast need not carry intervention authority.

Recursive address closure is reached only when every object and relation carrying the present conclusion has been typed, every claimed property transfer names its carrier relation, and further decomposition would no longer change the conclusion, route, boundary, falsifier, or outcome. This is sufficient resolution for the active claim, not exhaustive knowledge.

4.2 Context-conditioned object and admissible comparison

For system X , write the context-conditioned address

$$\mathcal{A}_X(t) = (S_X(t), I_X(t), P_X(t), O_X(t); \mathcal{C}_X(t)),$$

with

$$\mathcal{C}_X(t) = (\mathcal{E}_t, \mathcal{B}_t, \mathcal{H}_t, \ell_t, \mathcal{K}_t^{\text{supp}}).$$

$\mathcal{E}_t$ is the relevant environment, $\mathcal{B}_t$ the active boundary, $\mathcal{H}_t$ the retained history or lineage, ℓ_t the scale/resolution, and Ξ_t^{supp} the declared support or carrier conditions. Context is not a fifth SIPO layer. A context coordinate is retained only if changing it can change the object class, admissible dynamics, or load-bearing conclusion.

Let $\mathcal{M}_X$ be the declared comparison class. A deformation D is a lawful same-class comparison only if

$$D[\mathcal{A}_X] \in \mathcal{M}_X.$$

If a proposed perturbation changes the subject rather than one variable inside the subject, it cannot identify the claimed cause. Algebraic freedom to vary a coordinate is not evidence that the variation is physically or structurally admissible.

4.3 Measurement-relative observability

Let $\mathcal{Q}_O^{\text{adm}}$ be the declared family of admissible probes, with readouts

$$\mathcal{R}_{O,q} : \mathcal{X} \longrightarrow \mathcal{Y}_q.$$

Define measurement-relative equivalence by

$$x \sim_{\mathcal{O}} x' \iff \mathcal{R}_{O,q}(x) = \mathcal{R}_{O,q}(x') \quad \forall q \in \mathcal{Q}_O^{\text{adm}},$$

or by preregistered statistical indistinguishability for noisy observations. The resolved state is then an equivalence class in $\mathcal{X}/\sim_{\mathcal{O}}$. Measurement closure is therefore target-relative. If a prediction F_Y factors through the quotient on the feasible region,

$$F_Y = \overline{F}_Y \circ \pi_{\mathcal{O}},$$

then distinctions hidden by the readout are irrelevant to that target in the tested domain. This does not imply complete state resolution or universal unobservability.

4.4 Relational organisation and functional identity

Let R denote a candidate load-bearing relation and

$$\pi_R : X \longrightarrow Y$$

map lower states into a proposed relational state. Four claims must remain separate. **Token lineage** identifies which historical object or process is being followed. **Functional identity** names the relation that makes it this kind of whole for the declared role and environment. **Constitutive closure** states how that relation persists, is enforced, or restores itself under perturbation. **Law sufficiency** identifies which target properties can be propagated at the relational address without material unresolved lower-state dependence.

Two tokens can instantiate the same functional organisation, and one token can preserve organisation through carrier turnover. Neither fact erases provenance. Correlation, synchrony, common graph form, mutual information, or visual persistence is insufficient for constitutive closure.

For a candidate relation family $B_0 = \{b_1, \dots, b_n\}$ and declared capability measure K_L , a minimal sufficient load-bearing set is any

$$B^* \in \arg \min_{B \subseteq B_0} |B|$$

such that $K_L(B) \geq \kappa^*$ and removal of every member drops the system below the preregistered threshold. An ablation effect

$$\lambda_i(B) = K_L(B) - K_L(B \setminus \{b_i\})$$

tests whether relation b_i carries load in that background. Multiple minimal sets may reveal redundancy or alternative implementations rather than failure.

Functional identity across implementations additionally requires relation-preserving carrier substitution. If carrier m_i is replaced by $m_{(i)}$, while the declared relation is preserved, the relevant capability should remain within tolerance after resource and boundary differences are controlled. Failure means that the carrier is itself load-bearing, the relation description omitted a property, the substitution failed, or the original attribution was wrong.

More generally, for systems A and B with dynamics F_A, F_B , a proposed translation $\Phi : \mathcal{X}_A \rightarrow \mathcal{X}_B$ must preserve both the declared invariant and the work it performs:

$$d_K(K_B(\Phi x), K_A(x)) \leq \varepsilon_K, \quad d_B(F_B(\Phi x), \Phi(F_A(x))) \leq \varepsilon_F.$$

The target state must also satisfy the support conditions of the target comparison class. Same graph, symmetry, ratio, or language without dynamics and support preservation is analogy, not cross-implementation equivalence.

5. Admissibility before the active physical law

5.1 Compatibility and admissibility

Let a typed compatibility construction map the complete addressed state and context to an admissibility descriptor,

$$\Xi_t = \mathcal{Q}_t(\mathcal{U}_t, E_t).$$

Ξ_t may encode compatible support, boundary conditions, constraint satisfaction, operator-domain restrictions, viable controls, or a domain-specific combination. It is not presumed to be a scalar and does not automatically represent a new physical entity.

5.2 The active physical-law object

Represent the law active at the P-address as

$$\mathfrak{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}).$$

$\mathcal{D}_{P,t}$ is the addressed physical state, path, or operator domain; $\mathcal{T}_{P,t}$ is the transition rule or generator; and $\mathcal{W}_{P,t}$ is any domain-appropriate weighting, selection, or stochastic object. The decomposition prevents a change in allowed support from being hidden inside a fitted weighting and prevents a familiar generator from being applied outside its domain.

A law assembler is written

$$\mathfrak{C}_P : (\Xi_t, P_t, E_t) \mapsto \mathfrak{L}_{P,t}.$$

The cleanest special case is domain conditioning,

$$\mathcal{D}_{P,t} = \mathfrak{D}_P(\Xi_t),$$

with an established $\mathcal{T}$ and $\mathcal{W}$ acting on that domain. The general architecture permits a demonstrated relation to affect the domain, transition rule, weighting, or a typed combination. It does not presume that any such independent effect exists.

The typing dependency is

addressed admissibility $\rightarrow$ domain $\rightarrow$ domain-valid transition rule $\rightarrow$ domain-appropriate weighting or selection.

This is a logical order, not necessarily a temporal sequence. Some physical formalisms package transition and weighting inseparably; the law-object tuple permits that while still preventing a weighting from assigning probability to states outside its lawful support.

5.3 Propagation, measurement, and readdressing

Let the addressed physical state be

$$P_t = (\beta_t, p_t) \in \mathfrak{P} = \bigsqcup_{\beta \in A_P} (\{\beta\} \times \mathcal{P}_\beta).$$

After Ξ_t and $\mathfrak{L}_{P,t}$ have been assembled, let

$$P_{t+\Delta}^- \in \text{Propagate}_{P,\Delta}^{\mathfrak{L}_{P,t}}(P_t)$$

denote the pre-registration physical result. The propagation object may return a unique state, probability law, evolved quantum state or channel, or solution family according to the declared physical model. Measurement is then typed separately:

$$\text{Instrument}_{P \rightarrow (P,O)} : P_{t+\Delta}^- \mapsto (P_{t+\Delta}, O_{t+\Delta}).$$

The expanded capstone is

$$\text{Update}_{\text{SIPO},\Delta}(\mathcal{U}_t) = \text{Readdress}_{t+\Delta} \left[\mathcal{U}_t, \text{Instrument}_{P \rightarrow (P,O)} \left(\text{Propagate}_{P,\Delta}^{\mathfrak{C}_P(\mathcal{Q}_t(\mathcal{U}_t, E_t), P_t, E_t)}(P_t) \right) \right].$$

Its compact operator form is

$$\text{Update}_{\text{SIPO}} = \text{Readdress} \circ \text{Instrument}_{P \rightarrow (P,O)} \circ \text{Propagate}_P^{\mathfrak{L}_P}, \quad \mathfrak{L}_P = \mathfrak{C}_P \circ \mathcal{Q},$$

with the state and context arguments understood. Operationally,

addressed state and context $\rightarrow$ admissibility,
 $\rightarrow$ active physical-law object,
 $\rightarrow$ physical propagation,
 $\rightarrow$ measurement and record,
 $\rightarrow$ readdressing.

The instrument is typed $P \rightarrow (P, O)$ because a physical measurement can change the physical state while creating an observer-positioned record. Passive observation is the special case in which the instrument leaves the relevant P-state unchanged.

For realised path $\gamma_t : P_t \rightsquigarrow P_{t+\Delta}$, update the retained history by a typed append,

$$H_{t+\Delta} = H_t \oplus (\gamma_t, O_{t+\Delta}),$$

not by numerical addition. Readdressing then recomputes the contextual address, relation state, promoted variables, and next effective possibility object. Its minimum interpretation is

$$S_{t+\Delta}^{\text{eff}} = R_S(S_t^{\text{eff}}, I_{t+\Delta}, P_{t+\Delta}, O_{t+\Delta}, b_{t+\Delta}, H_{t+\Delta}).$$

The loop does not grant causal power to a record, imply that ambient S^* changed, or establish an independent information layer. Those are stronger claims with separate evidence burdens.

6. Adaptive state spaces and relational completion

6.1 Changing construction address

Learning, development, evolution, and architectural change may alter the variables, decoder, admissible operations, or update rule. To avoid treating all change as motion in a fixed space, use an addressed family

$$\mathfrak{X} = \bigsqcup_{\alpha \in A} (\{\alpha\} \times \mathcal{X}_\alpha).$$

An adaptive update is

$$\mathfrak{F}_t(\mathfrak{A}_t, u_t) = (\alpha_{t+1}, x_{t+1}) \in \mathfrak{X}.$$

If $\alpha_{t+1} \neq \alpha_t$, subtraction, distance, or state equivalence requires an explicit embedding, translation, quotient, or common observable. Without it, a numerical comparison is not merely noisy; it is undefined.

6.2 Typed relation-compatible region

Let

$$\mathcal{L}_t = \prod_i \mathcal{F}_t(U_i)$$

be an ambient assignment space and let $\mathfrak{E}_t = \{r_1, \dots, r_m\}$ be a family of typed relation scopes. A compatibility test χ_B selects

$$\mathfrak{C}_{B,t} = \{a \in \mathcal{L}_t : \chi_B(a) = 1\}.$$

Geometric overlap is only one relation class. Regulatory, temporal, contextual, system–environment, and semantic or functional relations require their own operational definitions.

6.3 Completion fibers and missing-address uncertainty

For a missing coordinate k and retained assignment a_{-k} , define

$$\mathfrak{F}_{k,t}(a_{-k}) = \mathfrak{C}_{B,t} \cap \pi_{-k}^{-1}(a_{-k}),$$

and the compatible completion set

$$\Omega_{k,t}^{\text{comp}}(a_{-k}) = \pi_k(\mathfrak{F}_{k,t}(a_{-k})).$$

The result distinguishes a unique completion, broad ambiguity, local incompatibility, and inconsistency elsewhere in the retained assignment. A compatible completion is not proof of actual physical occupancy.

6.4 Higher-scale readout and effective-object promotion

For a declared readout

$$R_{\ell \rightarrow L} : \mathfrak{C}_{B,\ell} \supseteq \text{Dom}(R) \longrightarrow \mathcal{Y}_L,$$

the lower-scale realisation class of macrostate y is

$$\mathcal{M}_y = \mathfrak{C}_{B,\ell} \cap R_{\ell \rightarrow L}^{-1}(y).$$

Lower-scale state can change while the declared macrostate remains fixed whenever the trajectory stays inside $\mathcal{M}_y$. The quotient induced by R remains a readout or coarse-graining until the candidate higher-scale object demonstrates persistence, interventionally load-bearing effects, new viable capability, predictive compression, and boundary or closure specificity.

6.5 Intrinsic organisation, extrinsic address, and path transport

For realised state $x_t \in \mathcal{X}_{\alpha_t}$, let

$$\kappa_t : \mathcal{X}_{\alpha_t} \longrightarrow \mathcal{K}_t$$

extract the intrinsic relation class claimed to remain constitutive, and let $b_t \in \mathcal{B}_t$ collect the relevant extrinsic address: position, orientation, external boundary, field configuration, neighbours, measurement frame, or system–environment coupling. The split is claim-relative, not a division into two substances.

For intrinsic class κ , extrinsic address b , and load-bearing history h , define the compatible physical realisation set

$$\mathcal{F}_P(\kappa; b, h) = \{p \in \mathfrak{P} : \chi_P(p; \kappa, b, h) = 1\}.$$

Intrinsic specification, compatible physical realisation, and actual physical occupancy are three different objects. A preserved intrinsic relation does not guarantee the same physical realisation in a different context, while loss of current P-occupancy does not erase a mathematical relation specification.

Where transport through context space is meaningful, let path $\gamma : b_0 \rightarrow b_1$ induce

$$T_\gamma : \mathcal{F}_P(\kappa_0; b_0, h_0) \supseteq \mathcal{D}_\gamma \longrightarrow \mathcal{F}_P(\kappa_1; b_1, h_1).$$

The transport may be path-dependent: two paths with the same endpoints may yield distinguishable results. “Holonomy” is reserved for implementations that genuinely provide the relevant connection/bundle structure; otherwise “loop transport” is the correct term [31,32]. Same endpoint is not a continuity proof when the tested state, phase, identity, or admissibility is history-dependent.

6.6 Physical persistence, relational addressability, and identity through scale change

A lower-scale variable can remain physically instantiated while losing independent deployment inside the newly organised whole. To represent this, let σ_t be an operational individuality address containing the construction address, active boundary, constitutive system–environment coupling class, maintenance relations, lineage/history, and within-address state-equivalence rule.

Three relations must not be substituted for one another:

$$\text{within-address state equivalence} \neq \text{address-class parity} \neq \text{cross-address identity continuity}.$$

If the individuality address changes class, a claim that “the same object lost a degree of freedom” requires a continuation map and a declared identity-bearing invariant. If identity depends on recent trajectory, endpoint transport is insufficient: the minimum history horizon must be determined by truncation, temporal shuffle, matched-present/different-history, and held-out tests.

The architecture therefore distinguishes:

1. same-address-class restriction;
2. readdressed continuation supported by a lawful state/history map;
3. incorporation, replacement, branching, or unresolved identity transition.

These are operational system-identity classes. They do not establish consciousness, personhood, a metaphysical soul, or persistence beyond physical support.

6.7 Scale transition, capability, and hidden cost

Let $\mathcal{F}_\ell$ be a lower-scale feasible set, $\mathcal{K}_L$ a higher-scale viable capability set, and $\mathbf{c}$ a typed cost vector containing only quantities actually measured, such as energy, material, time, error burden, or maintenance demand. Report the transition first as

$$\mathbf{Q}_t = (\Delta\mu_\ell, \Delta\nu_L, \Delta\mathbf{c}),$$

where μ_ℓ and ν_L are measures on different declared objects. The tuple is not a conservation law and should not be scalarised without lawful normalisation.

An enabling constraint can produce $\Delta\mu_\ell < 0$ and $\Delta\nu_L > 0$; overconstraint can make both negative. Neither pattern fixes the sign of cost. Persistent relational mismatch may be accumulated as an exposure only when successive residuals are comparable; it becomes “cost” only when it predicts or causally changes a separately measured cost

coordinate. This is the negative-space guard: improved visible performance does not license omission of transferred burden, maintenance demand, lost options, or delayed failure.

A candidate higher-scale object earns promotion only when the following conjunction is supported under strong alternatives:

1. persistence over the timescale relevant to the higher description;
2. predicted deformation when internal load-bearing relations are changed;
3. at least one viable capability unavailable to a matched independent-part baseline under the same resource envelope;
4. predictive compression on held-out outcomes without material omitted-state failure;
5. a reproducible boundary or closure specificity not shared by arbitrary regroupings.

Promotion changes the permitted model address. It does not erase lower-scale physics or prove autonomous higher-level causation.

7. Interfaces, law descent, and cross-scale recovery

7.1 Dynamic-interface promotion

Let $\mathcal{J}_{AB,t}$ be a trackable interface state between addressed systems A_t and B_t . The interface remains bundled context unless three conditions hold: it is operationally definable, its controlled deformation changes a preregistered downstream object, and the realised transition updates it in a way that changes the next cycle. The recursive signature is

$$\mathcal{J}_{AB,t} \longrightarrow (A_{t+\Delta}, B_{t+\Delta}) \longrightarrow \mathcal{J}_{AB,t+\Delta} \longrightarrow \Xi_{t+\Delta}.$$

Passing this gate promotes the interface into explicit state representation, not into a new ontology. Physical work required to create, move, modulate, or maintain an interface remains in the physical energy ledger.

If the coupled object later passes the independent scale-promotion gate, a boundary at scale L may become an internal relation of the promoted whole at $L + 1$. This is a change of role with address, not creation of a fifth SIPO layer. If explicit interface state adds no prospective discrimination, the simpler bundled context controls.

7.2 Property-relative law descent

Let

$$\pi_R : X \longrightarrow Y$$

map lower-address states to a candidate higher relational state under relation R . For matched deterministic lower dynamics F_X , strong whole-state descent requires

$$\pi_R(x) = \pi_R(x') \implies \pi_R(F_X(x)) = \pi_R(F_X(x'))$$

within a declared environment, intervention class, timescale, and tolerance. A claim about one target property q tests descent for q rather than silently certifying all properties of Y .

For stochastic lower dynamics with kernel K_X^Δ , strong approximate descent requires pushed-forward transition laws for representatives of the same relational state to agree within a preregistered tolerance:

$$\epsilon_{\text{strong}} = \sup_{x \sim_R x'} d(\pi_{R*} K_X^\Delta(\cdot \mid x, u, e), \pi_{R*} K_X^\Delta(\cdot \mid x', u, e)) \leq \epsilon_*.$$

Where representative-independent descent fails, an ensemble law may still be supported for an independently justified preparation distribution. Where one-step closure fails, a bounded higher-level history can be added and retested. Unlimited history cannot be used to hide material unresolved lower-state dependence.

The supported class must be named: exact, approximate, ensemble or conditional, history-augmented, or coupled multiscale. Predictive sufficiency does not imply interventional sufficiency. If the higher address is used for control or causal authority, it must survive the declared intervention class.

7.3 Closure and law sufficiency are different boundaries

Let M_C measure margin to loss of the constitutive closure relation and $M_D(q)$ margin to loss of higher-address dynamical sufficiency for property q . The architecture permits

$$M_D(q) < M_C,$$

so that a higher-level law for q can fail before the organised whole loses closure. No universal ordering is assumed.

Readdressing is therefore bidirectional:

$$\text{lower address} \nleftrightarrow \text{higher/effective address} \nleftrightarrow \text{coupled multiscale address}.$$

The active law is assembled at the coarsest address that has actually earned sufficiency for the declared property and regime. If preparation dependence, intervention, fracture, timescale, or hidden representative state destroys that sufficiency, the model must readdress downward or multiscale without denying that the higher functional object may remain real. This is the strongest current formulation of **address before law ownership**.

7.4 Local performance and whole-system condition

For a lower-address performance variable J_ℓ , retain the tuple

$$\Delta_{\ell \rightarrow h} = (\Delta J_\ell, \Delta M_C, \Delta M_D(q)).$$

The entries may have different units and must not be collapsed without a justified normalisation. Local improvement does not inherit upward. An antagonistic regime with $\Delta J_\ell > 0$ and $\Delta M_C < 0$ is possible, but aligned, neutral/decoupled, and mixed/property-relative cases are equally admissible. The sign must be measured at each address; no universal “local gain causes collapse” law is retained.

7.5 Conditional recoverability

Let $\mathcal{C}_R \subseteq Y$ be a target closure class, with admissible controls U , environment class E , route constraints, and horizon H . Define the recovery indicator

$$\text{Rec}_H(x_t; \mathcal{C}_R, U, E) = \begin{cases} 1, & \text{if an admissible route reaches } \mathcal{C}_R \text{ within } H, \\ 0, & \text{otherwise.} \end{cases}$$

Recovery is target-relative reachability, not a promise that the degradation path can be reversed. A different lower-scale realisation may recover the same functional class. Apparent hysteresis must first be tested against measurable omitted state and bounded state augmentation.

Three transition boundaries are therefore tracked separately:

1. loss of property-specific higher-law sufficiency;
2. loss of constitutive closure;
3. loss of recoverability to the declared target under the declared controls, environment, horizon, and admissible route class.

A system may have $M_D(q) < 0$ while $M_C > 0$ and $\text{Rec}_H = 1$: the macro law for q has failed, the organised whole persists, and a recovery route remains available. Conversely, recognisable structure can persist after a particular recovery target is no longer reachable. Reduced predictability is not collapse, and closure loss is not by itself irreversibility.

7.6 Asynchronous transition and history guard

Layer or scale variables need not readdress simultaneously. A lead-lag is admissible only when a typed transition model predicts which variable moves first, the downstream resolution time, and the bounds within which the mismatch should close. Otherwise the mismatch is evidence against the proposed route.

Similarly, asymmetric return paths do not by themselves establish irreducible memory. Before history is promoted to a constitutive coordinate, ordinary drift, physical memory, hysteresis, instrument state, and bounded state augmentation must be tested. If an omitted measurable state restores predictive closure, that state belongs in the address.

8. Foundational physics burden

The architecture becomes a physical theory only when $\mathcal{Q}$, $\mathcal{C}_P$, and $\mathcal{L}_P$ are instantiated in concrete regimes. The following are acceptance requirements rather than results already obtained.

8.1 Entanglement and Bell structure

For a bipartite experiment with settings a, b and outcomes $A, B \in \{-1, +1\}$, any MKUFT model must produce a well-defined joint distribution $P(A, B \mid a, b)$ and the correlations

$$E(a, b) = \sum_{A, B=\pm 1} AB P(A, B | a, b).$$

For CHSH settings, define

$$S_{\text{CHSH}} = E(a, b) + E(a, b') + E(a', b) - E(a', b').$$

Local hidden-variable models satisfy $|S_{\text{CHSH}}| \leq 2$, while quantum theory permits values up to $2\sqrt{2}$ [2,3]. A relational label is inadequate unless the proposed law object recovers the tested quantum correlations without reintroducing an excluded local-hidden-variable mechanism.

8.2 No-signalling

The observed marginal at one wing must not depend on the distant setting:

$$\sum_B P(A, B | a, b) = \sum_B P(A, B | a, b')$$

and symmetrically for B [4]. Any operational signalling introduced by the deeper architecture falsifies the proposed entanglement branch.

8.3 Quantum and quantum-field-theory recovery

A successful instantiation must recover the Born rule, unitary or otherwise experimentally adequate dynamics, composition and tensor-product structure where applicable, identical-particle statistics, relativistic causal structure, and the tested quantum-field-theory limits. An architecture that only reproduces a qualitative language of possibility and constraint has not met this burden.

8.4 Gravitational recovery

The gravitational branch must recover Einstein dynamics or an experimentally equivalent limit, including the tested weak-field, post-Newtonian, lensing, timing, wave, and conservation behaviour. Constraint and boundary structures are already part of general relativity and effective field theory [6,7]. Re-describing them as admissibility does not constitute a new gravitational mechanism.

8.5 Strongest fair physical null

The null is not “no relation.” It is the strongest adequate physical-state, history, boundary, environment, and measurement-context model. If that model determines the same law object and predictions, the relation is physically realised within P and no independent $I \rightarrow P$ claim is earned.

9. Mathematical scaffold and equation status

9.1 Independent-content test

Let X_t be the measured physical and environmental state, H_t the measured history, I_t a candidate information variable, and $Y_{t+\Delta}$ a preregistered physical outcome. The physical-only null is

$$H_0 : Y_{t+\Delta} \perp\!\!\!\perp I_t | X_t, H_t.$$

An information variable earns independent content only if it adds held-out predictive or interventional value, survives relation-preserving re-encoding, and cannot be reduced to an adequate function of the controlled physical state and history.

The present architecture distinguishes four evidential levels:

- **Level 0 – P-only:** the complete law object is determined by an adequate physical state, history, boundary/environment, and measurement-context model.
- **Level 1 – relational compression:** I_t predicts or compresses the law object efficiently but is recoverable from P.
- **Level 2 – P-realised load-bearing relation:** relation-preserving versus relation-disrupting intervention changes $\mathcal{D}$, $\mathcal{T}$, or $\mathcal{W}$, and an adequate P-description predicts the change.
- **Level 3 – independent I candidate:** a separately specified I_t adds held-out or interventional prediction of the law object after the strongest adequate P account has been controlled.

Only Level 3 begins to support independent $I \rightarrow P$ dynamics, and it remains conditional on conservation, no-signalling, ordinary-limit recovery, and a discriminating replication. A non-zero conditional association is not automatically causal.

Operational semantic load-bearingness is likewise defined by consequence, not by human meaningfulness. Preserving versus disrupting a proposed relation must change a preregistered future observable, action distribution, viability condition, admissible support, or other system-relevant outcome while lower-order statistics and carrier details are controlled as far as practical [33]. If it does not, the relation is not load-bearing for that claim.

9.2 Supersession of the historical event-weight product

The historical product of physical, informational, and observer factors is not reproduced as a current flagship equation. Merely replacing multiplication by an integral and normalisation would make the expression mathematically legal, but it would not derive the factors, identify a coupling, or establish an observer-conditioned physical effect.

Any future event-weight proposal belongs inside the domain-appropriate $\mathcal{W}_{P,t}$ component of a concrete law object. It must define its event space, physical state, measure, positivity, normalisation, units, boundary conditions, and measurement route; recover the accepted physical distribution; and demonstrate an independently specified relational variable beyond the strongest adequate P-model. Until those burdens are met, the historical ansatz is provenance, not a live physical law.

9.3 Path and cost scaffold

For a trajectory γ in one declared state space,

$$C[\gamma] = \int_{\gamma} \lambda(x) ds.$$

The path element, cost density, and units must be specified. A Gibbs-like path model requires a dimensionless exponent:

$$P(B | A) = \frac{1}{Z_A} \sum_{\gamma \in \Gamma(A \rightarrow B)} \exp[-\beta C[\gamma]],$$

where $\beta C[\gamma]$ is dimensionless. Path multiplicity and cost are candidate predictors, not a universal definition of probability.

9.4 Equation-status classes

- **E0 – notation or bookkeeping:** typed addresses, tuples, projections, and comparison labels.
- **E1 – research scaffold:** mathematically legal candidate forms without physical derivation or validation.
- **E2 – operational statistical model:** measured variables, frozen estimation, uncertainty, and explicit baselines.
- **E3 – replicated empirical model:** held-out and independently replicated support for the tested relation.
- **E4 – derived physical mechanism:** defined physical states, couplings, symmetries, conservation conditions, accepted-limit recovery, and a discriminating quantitative prediction.

No current foundational MKUFT equation is classified as E4.

10. Experimental and falsification programme

10.1 Evidence progression

The public experimental route is

calibration $\longrightarrow$ reconnaissance $\longrightarrow$ locked signature $\longrightarrow$ strongest fair null,
 $\longrightarrow$ deformation and ablation $\longrightarrow$ held-out prediction $\longrightarrow$ independent replication.

Exploratory significance does not promote a branch. Confirmatory studies require preregistration, power and precision analysis, frozen exclusions, multiplicity control, complete reporting, and independent replication.

10.2 Priority lanes

The near-term operational lane tests typed active traversal and load-bearing dependency against complete-history replay, independent, one-way, reset, relation-scrambled, and serious alternative-pairing controls [19]. That programme can test architectural claims without assuming speculative physics.

The foundational lane instantiates the addressed law architecture and attempts quantum and gravitational recovery. Its success criterion is a correct derivation plus a prospective discriminator; qualitative fit is insufficient.

Controlled physical-residual studies may use boundary-conditioned optical, acoustic, dielectric, fluid, plasma, or material systems. They require a defined apparatus, ordinary baseline, residual equation, parameter sweep, and predeclared response surface.

RNG, remote-information, collective-correlation, and observer-state assays remain reconnaissance or falsification branches. A generic anomaly would not establish a unified field, a substrate, or observer-driven physical causation.

10.3 Strongest-fair-null and deformation design

For a declared reciprocal system $A \leftrightarrow B$ and scalar performance measure F , the controlling null is the strongest commensurable result among competent independent, combined-independent, one-way-assisted, alternative-pairing, and complete-history-replay conditions under a matched resource envelope:

$$F_{\text{null}}^* = \max\{F_A^*, F_B^*, F_{\text{ind}}^*, F_{A \rightarrow B}^*, F_{B \rightarrow A}^*, F_{\text{replay}}^*\}.$$

The live reciprocal gain is $G_{\text{syn}}^* = F_{AB} - F_{\text{null}}^*$. Relationship specificity additionally requires an advantage over strong alternative pairings, while path dependence requires an advantage over a complete explicit-history replay on held-out path-sensitive tasks. If replay matches the live condition, the result may still demonstrate valuable transferable architecture, but not live path-specific gain.

Vector outcomes remain vectors unless a scalarisation or Pareto rule is preregistered. Time, compute, information access, preparation, revision opportunities, history-bundle completeness, evaluator exposure, and scoring direction must be matched or explicitly modelled.

Across relational and scale-transition branches, the minimum discriminating deformation set is:

1. remove each proposed load-bearing relation singly and in relevant coalitions;
2. preserve the relation while substituting a materially different carrier;
3. preserve lower-order statistics while scrambling the higher-order relation;
4. hold intrinsic organisation fixed while changing extrinsic address or boundary;
5. compare same endpoint reached by different controlled paths;
6. perturb lower states inside and across a proposed macro realisation class;
7. test whether an apparent interface variable adds prospective value after ordinary boundary and history terms;
8. compare the promoted object against serious alternative coarse-grainings;
9. test recovery to a declared target under declared controls and horizon;
10. rerun every claimed residual against the strongest ordinary physical model.

10.4 Branch-level failure conditions

A branch is reduced or rejected if:

- its variables cannot be operationally defined;
- direct comparison across changed addresses lacks a lawful map;
- a proposed probability is negative or non-normalisable;
- a dimensionful quantity enters an exponential without a declared scale;
- the ordinary physical limit cannot be recovered;
- the strongest fair null predicts equally well or better;
- a named relation can be ablated or scrambled without the predicted deformation;
- held-out predictions fail or independent replications converge to the null;
- post-hoc flexibility is required to preserve the claim;
- evidence from one layer is used to rescue failure at another.
- a relation-specific effect disappears under carrier substitution or strong alternative representations;
- a claimed scale transition is only analyst coarse-graining and produces no endogenous change in accessible relations;
- claimed hidden cost is inferred from mismatch without a separately measured cost coordinate;
- a same-self comparison crosses an individuality-address change without a continuation map;
- a recovery or irreversibility claim lacks a target class, route constraints, controls, environment, or time horizon.

10.5 Framework-level reduction

Failure of one branch does not validate another. If no proposed cross-layer variable adds independent content, MKUFT may reduce to a useful typed methodology for state representation, model selection, and cross-scale auditing. If the capstone cannot be instantiated consistently, even that architectural claim must be narrowed. If quantum and gravitational recovery fail, the foundational unified-field claim fails regardless of successes in non-foundational applications.

11. Prior art and residual contribution

MKUFT does not claim priority over measure spaces, Hilbert spaces, graph paths, conditional probability, Markov kernels, linear response, coarse-graining, lumpability, causal abstraction, organisational closure, enablement, viability, reachability, hysteresis, quantum measurement instruments, general-relativistic constraints, or boundary-conditioned dynamics.

Close scientific neighbours include operational and informational reconstructions of quantum theory [8,9], invariant-set approaches [10], emergent-gravity and holographic programmes [11–14], constraint-closure and changing-possibility-space work [15–18], causal abstraction and macro-level efficacy [22–25], and memory-aware reduced dynamics [26,27]. These fields remove any broad claim that admissibility, emergence, constraint, or higher-level law is new in isolation.

After that subtraction, the candidate MKUFT contribution is the conjunction of:

- a typed S–I–P–O address discipline;
- explicit construction of admissibility before the active law object;
- separation of physical domain, transition rule, and weighting;
- a propagation–instrument–readdressing loop with retained history;
- explicit guards against comparison across changed construction addresses;
- interface promotion only after trackability, prospective load, and recursive back-action;
- property-relative higher-address law sufficiency;
- separate margins for closure, law descent, and recoverability;
- evidence-triggered bidirectional readdressing;
- strongest-fair-null and controlled-deformation governance.

The strongest defensible description is an original operational synthesis or meta-principle for deciding which addressed object may own a law for a declared property. Its novelty remains defeasible by closer prior art and its scientific value remains dependent on prospective use.

12. Discussion

The principal correction from historical MKUFT is not cosmetic. The early publication placed a speculative observer-modulated probability product near the centre of the framework. The current publication removes that expression from the governing equations; any mathematically repaired descendant remains an optional E1 branch in the supporting canon. The addressed law-selection factorisation is now primary. This prevents observer or information language from carrying physical force by assertion.

The second correction is cross-scale. A higher-level object, a higher-level predictive variable, constitutive closure, law sufficiency, and recoverability are different objects. Treating them as one produces false emergence claims and false collapse claims. The property-relative descent test and recovery route make those distinctions operational.

The third correction is scientific. The architecture is not allowed to substitute coherence for derivation. Bell compatibility, no-signalling, quantum and gravitational recovery, dimensional consistency, energy accounting, and strong alternatives remain external constraints on the programme.

The framework may therefore fail in several coherent ways. It may survive as an audit architecture while failing as foundational physics. A particular state-space or recovery model may prove useful without establishing an independent information layer. A relational physical mechanism may close entirely within P. Those are reductions, not evasions.

13. Conclusion

MKUFT now has a closed architectural drivetrain:

possibility $\longrightarrow$ typed relation $\longrightarrow$ admissibility $\longrightarrow$ active physical law,
 $\longrightarrow$ propagation and measurement $\longrightarrow$ readdressed possibility.

The drivetrain is extended across scale by explicit comparison, effective-object promotion, interface state, property-relative law descent, separate closure and law-sufficiency margins, and conditional recovery. This is sufficient to state a coherent research architecture. It is not sufficient to claim completed physical unification.

The decisive next work is concrete instantiation. A successful foundational branch must recover the established quantum and gravitational regimes and produce at least one robust discriminator beyond the strongest existing account. Until then, MKUFT should be read as a pressure-testable relational architecture for physical-law selection and cross-scale dynamics—not as accepted physics, and not as a claim that an attractive formal pattern has already become a law of nature.

Appendix A. Notation and status map

- $S = (\Omega, \Sigma, \mu)$: ambient possibility scaffold; E0/E1 depending on use.
- S_t^{eff} : current reachable or addressable possibility object; E0 until operationalised.
- $I, \mathcal{I}, i$: information address, information-function space, and one information structure.
- P_t, O_t : physical state and observer-positioned record or bounded state.
- b_t, H_t, E_t : contextual address, retained history, and measured environment.
- $\mathcal{U}_t$: complete addressed state; E0 bookkeeping.
- $\Xi_t = \mathcal{Q}_t(\mathcal{U}_t, E_t)$: admissibility descriptor; E1 until domain-specific.
- $\mathcal{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t})$: active physical-law object; E0/E1 architecture.
- $\mathcal{C}_P$: law assembler; E1 until instantiated.
- $\mathcal{C}_{B,t}$: typed relation-compatible region.
- $\mathcal{M}_y$: lower-scale realisation class of macrostate y .
- π_B : declared lower-to-higher relation or coarse-graining map.
- $B^*, \lambda_i(B)$: minimal sufficient load-bearing relation set and relation-ablation effect.
- $\kappa, b, \mathcal{F}_P(\kappa; b, h)$: intrinsic relation class, extrinsic address, and compatible P-realisation set.
- σ_t : operational individuality address; distinct from within-address state equivalence and cross-address identity continuity.
- $M_C, M_D(q)$: closure and property-specific law-sufficiency margins.
- Rec_H : conditional recovery reachability indicator.
- $\mathbf{Q}_t = (\Delta\mu_t, \Delta\nu_L, \Delta c)$: separate lower feasible-set, higher capability, and typed cost changes.

Appendix B. Minimum acceptance gates for a physical instantiation

1. Define the exact physical state space, variables, units, boundary conditions, and measurement route.
2. Define $\mathcal{Q}$ and $\mathcal{C}_P$ without retrospective fitting to the target outcome.
3. Demonstrate positivity, normalisation, operator-domain validity, and dimensional consistency where applicable.
4. Recover the relevant accepted physical limit within a declared norm and tolerance.
5. For an entanglement branch, derive the tested joint probabilities, Bell-compatible correlations, and no-signalling marginals.
6. For a gravitational branch, recover the tested general-relativistic regime or a quantitatively equivalent limit.
7. State the strongest fair alternative model and lock a prospective discriminator.
8. Ablate or scramble the proposed load-bearing relation and predict the deformation before observing it.
9. Separate exploratory fitting from held-out testing and independent replication.
10. Reduce or reject the branch when the acceptance conditions fail.

References

1. Einstein, A., Podolsky, B., & Rosen, N. (1935). Can Quantum-Mechanical Description of Physical Reality Be Considered Complete? *Physical Review*, 47, 777–780. DOI: [10.1103/PhysRev.47.777](https://doi.org/10.1103/PhysRev.47.777).
2. Bell, J. S. (1964). On the Einstein Podolsky Rosen Paradox. *Physics Physique Fizika*, 1, 195–200. DOI: [10.1103/PhysicsPhysiqueFizika.1.195](https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195).
3. Clauser, J. F., Horne, M. A., Shimony, A., & Holt, R. A. (1969). Proposed Experiment to Test Local Hidden-Variable Theories. *Physical Review Letters*, 23, 880–884. DOI: [10.1103/PhysRevLett.23.880](https://doi.org/10.1103/PhysRevLett.23.880).
4. Simon, C., Bužek, V., & Gisin, N. (2001). No-Signaling Condition and Quantum Dynamics. *Physical Review Letters*, 87, 170405. DOI: [10.1103/PhysRevLett.87.170405](https://doi.org/10.1103/PhysRevLett.87.170405).
5. Hensen, B., Bernien, H., Dréau, A. E., et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature*, 526, 682–686. DOI: [10.1038/nature15759](https://doi.org/10.1038/nature15759).
6. Arnowitt, R., Deser, S., & Misner, C. W. (1962/2004). The Dynamics of General Relativity. arXiv: gr-qc/0405109.
7. Donoghue, J. F. (1994). General relativity as an effective field theory: The leading quantum corrections. *Physical Review D*, 50, 3874–3888. DOI: [10.1103/PhysRevD.50.3874](https://doi.org/10.1103/PhysRevD.50.3874).
8. Hardy, L. (2001). Quantum Theory From Five Reasonable Axioms. arXiv: quant-ph/0101012.
9. Chiribella, G., D'Ariano, G. M., & Perinotti, P. (2011). Informational derivation of quantum theory. *Physical Review A*, 84, 012311. DOI: [10.1103/PhysRevA.84.012311](https://doi.org/10.1103/PhysRevA.84.012311).
10. Palmer, T. N. (2009). The Invariant Set Postulate: a new geometric framework for the foundations of quantum theory and the role played by gravity. *Proceedings of the Royal Society A*, 465, 3165–3185. DOI: [10.1098/rspa.2009.0080](https://doi.org/10.1098/rspa.2009.0080).
11. Jacobson, T. (1995). Thermodynamics of Spacetime: The Einstein Equation of State. *Physical Review Letters*, 75, 1260–1263. DOI: [10.1103/PhysRevLett.75.1260](https://doi.org/10.1103/PhysRevLett.75.1260).
12. Maldacena, J. M. (1998). The Large N Limit of Superconformal Field Theories and Supergravity. *Advances in Theoretical and Mathematical Physics*, 2, 231–252. DOI: [10.4310/ATMP.1998.v2.n2.a1](https://doi.org/10.4310/ATMP.1998.v2.n2.a1).
13. Ryu, S., & Takayanagi, T. (2006). Holographic Derivation of Entanglement Entropy from the anti-de Sitter Space/Conformal Field Theory Correspondence. *Physical Review Letters*, 96, 181602. DOI: [10.1103/PhysRevLett.96.181602](https://doi.org/10.1103/PhysRevLett.96.181602).
14. Van Raamsdonk, M. (2010). Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42, 2323–2329. DOI: [10.1007/s10714-010-1034-0](https://doi.org/10.1007/s10714-010-1034-0).
15. Mossio, M., & Moreno, A. (2010). Organisational closure in biological organisms. *History and Philosophy of the Life Sciences*, 32, 269–288.
16. Montévil, M., & Mossio, M. (2015). Biological organisation as closure of constraints. *Journal of Theoretical Biology*, 372, 179–191. DOI: [10.1016/j.jtbi.2015.02.029](https://doi.org/10.1016/j.jtbi.2015.02.029).
17. Pattee, H. H. (2001). The physics of symbols: bridging the epistemic cut. *Biosystems*, 60, 5–21. DOI: [10.1016/S0303-2647\(01\)00104-6](https://doi.org/10.1016/S0303-2647(01)00104-6).
18. Longo, G., Montévil, M., & Kauffman, S. (2012). No entailing laws, but enablement in the evolution of the biosphere. *GECCO '12 Companion*, 1379–1392. DOI: [10.1145/2330784.2330946](https://doi.org/10.1145/2330784.2330946).

19. McLaughlin, M. C. (2026). Active Traversal and Load-Bearing Dependency in Typed Knowledge Architectures: A Matched-Control Evaluation Protocol for AI Systems. Zenodo. DOI: [10.5281/zenodo.21341521](https://doi.org/10.5281/zenodo.21341521).
20. McLaughlin, M. C. (2026). Layer Before Law: An Addressed Architecture for Relational Admissibility, Effective Law Selection, and Quantum-Gravitational Recovery. Zenodo. DOI: [10.5281/zenodo.21971270](https://doi.org/10.5281/zenodo.21971270).
21. McLaughlin, M. C. (2026). Recursive Constraint Closure and Reachable-State Geometry: An Addressed Formalism for Adaptive Systems with Changing Possibility Spaces. Zenodo. DOI: [10.5281/zenodo.21971425](https://doi.org/10.5281/zenodo.21971425).
22. Rubenstein, P. K., Weichwald, S., Bongers, S., et al. (2017). Causal Consistency of Structural Equation Models. *Proceedings of UAI 2017*. arXiv: 1707.00819.
23. Beckers, S., & Halpern, J. Y. (2019). Abstracting Causal Models. *AAAI*, 33, 2678–2685. DOI: [10.1609/aaai.v33i01.33012678](https://doi.org/10.1609/aaai.v33i01.33012678).
24. Beckers, S., Eberhardt, F., & Halpern, J. Y. (2020). Approximate Causal Abstractions. *Proceedings of Machine Learning Research*, 115, 606–615. arXiv: 1906.11583.
25. Hoel, E. P., Albantakis, L., & Tononi, G. (2013). Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110, 19790–19795. DOI: [10.1073/pnas.1314922110](https://doi.org/10.1073/pnas.1314922110).
26. Li, Z., Bian, X., Li, X., & Karniadakis, G. E. (2015). Incorporation of memory effects in coarse-grained modeling via the Mori–Zwanzig formalism. *Journal of Chemical Physics*, 143, 243128. DOI: [10.1063/1.4935490](https://doi.org/10.1063/1.4935490).
27. Gouasmi, A., Parish, E. J., & Duraisamy, K. (2017). A priori estimation of memory effects in reduced-order models of nonlinear systems using the Mori–Zwanzig formalism. *Proceedings of the Royal Society A*, 473, 20170385. DOI: [10.1098/rspa.2017.0385](https://doi.org/10.1098/rspa.2017.0385).
28. Ozawa, M. (2001). Quantum Measurement, Information, and Completely Positive Maps. arXiv: quant-ph/0107090.
29. Facchi, P., Garnero, G., & Ligabò, M. (2018). Self-adjoint extensions and unitary operators on the boundary. *Letters in Mathematical Physics*, 108, 195–212. DOI: [10.1007/s11005-017-1001-8](https://doi.org/10.1007/s11005-017-1001-8).
30. McLaughlin, M. C. (2025). Unified Field Theory. McLaughlin–Kairos. MKUFT (Version v1). Zenodo. DOI: [10.5281/zenodo.17780566](https://doi.org/10.5281/zenodo.17780566).
31. Simon, B. (1983). Holonomy, the Quantum Adiabatic Theorem, and Berry's Phase. *Physical Review Letters*, 51, 2167–2170. DOI: [10.1103/PhysRevLett.51.2167](https://doi.org/10.1103/PhysRevLett.51.2167).
32. Berry, M. V. (1984). Quantal phase factors accompanying adiabatic changes. *Proceedings of the Royal Society A*, 392, 45–57. DOI: [10.1098/rspa.1984.0023](https://doi.org/10.1098/rspa.1984.0023).
33. Kolchinsky, A., & Wolpert, D. H. (2018). Semantic information, autonomous agency and non-equilibrium statistical physics. *Interface Focus*, 8, 20180041. DOI: [10.1098/rsfs.2018.0041](https://doi.org/10.1098/rsfs.2018.0041).

Appendix C. Public source provenance

This paper was reconciled against the public MKUFT repository, including the integrated master spine and the public core, mathematical, experimental, falsification, claim-discipline, Layer-Before-Law, adaptive-state, SIPO-capstone, interface, law-descent, recoverability, novelty-audit, and scientific-reference modules. The audited public-canon snapshot is Git commit [45a07d31b0b28ce3e5e607d388742dbb3c4e73e2](https://git.mkuft.org/commit/45a07d31b0b28ce3e5e607d388742dbb3c4e73e2).

The standalone papers *Layer Before Law* (DOI [10.5281/zenodo.21971270](https://doi.org/10.5281/zenodo.21971270)) and *Recursive Constraint Closure and Reachable-State Geometry* (DOI [10.5281/zenodo.21971425](https://doi.org/10.5281/zenodo.21971425)) remain distinct frozen objects. The historical lineage paper remains DOI [10.5281/zenodo.17780566](https://doi.org/10.5281/zenodo.17780566). This publication integrates only the dependencies required for the current principal MKUFT statement; it does not silently replace those objects or inherit every supporting hypothesis. The deposited publication package must identify the exact frozen manuscript and PDF by checksum. Later repository changes do not alter the deposited object.